Enterprise Churn & Revenue Risk Intelligence System

Machine Learning Project Report

Objective: analyse customer churn patterns, train classification models, compare their performance, and evaluate a recall-focused Random Forest decision threshold.

Dataset: IBM Telco Customer Churn sample dataset

Dataset size: 7,043 records and 21 original columns

Selected model: Random Forest, threshold 0.38

Prepared from the outputs generated in the Google Colab workflow.

1. Project Objective

The project explores customer churn classification using three machine learning algorithms. The workflow includes data auditing, cleaning, exploratory analysis, preprocessing, model training, held-out test evaluation, feature importance analysis, and threshold selection for churn detection.

2. Data Preparation

The original dataset contains 7,043 rows and 21 columns. The customerID column was excluded from model features. Blank TotalCharges values associated with zero-tenure records were converted to numeric zero. The target variable was encoded as No = 0 and Yes = 1. The data was split into 5,634 training records and 1,409 test records using stratification.

3. Model Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	ROC-AUC
Gradient Boosting	79.77	65.95	49.20	56.36	0.8428
Logistic Regression	80.55	65.72	55.88	60.40	0.8421
Random Forest	77.93	57.63	63.64	60.48	0.8340

4. Final Test Evaluation

Metric	Test result
Accuracy	74.73%
Precision	51.64%
Recall	75.94%
F1-score	61.47%
ROC-AUC	0.8340

5. Feature Importance

The Random Forest model's top features from the fitted pipeline are listed below. Feature importance measures contribution to the model's predictive splits; it does not establish causation.

Feature	Importance
numeric__tenure	0.12792
numeric__TotalCharges	0.12481
numeric__MonthlyCharges	0.09977
categorical__Contract_Month-to-month	0.09444
categorical__OnlineSecurity_No	0.04787
categorical__Contract_Two year	0.04522
categorical__TechSupport_No	0.03891
categorical__InternetService_Fiber optic	0.03656
categorical__PaymentMethod_Electronic check	0.03143
categorical__OnlineBackup_No	0.01746

6. Threshold and Business Trade-off

A decision threshold of 0.38 was selected using five-fold stratified cross-validation on the training data to maximize F1-score. On the held-out test set, this threshold achieved 75.94% churn recall and 51.64% churn precision, with 74.73% accuracy. The threshold identifies more churn cases but also produces false positives, so the appropriate threshold depends on retention costs and business priorities.

7. Limitations

This is a sample dataset, not live customer data from an operating telecom company. Results may not generalize to other businesses or future customer populations. Model predictions are estimates, not guarantees of future behaviour. Feature importance is not causal evidence. Threshold selection should be validated against real retention costs before deployment.

8. Conclusion

The project demonstrates an end-to-end churn classification workflow and a comparison of Logistic Regression, Random Forest, and Gradient Boosting. Random Forest was selected for the recall-focused threshold experiment. The test results show a trade-off between identifying churn customers and generating false-positive alerts. Further work could include cost-based threshold optimization, external validation, and monitoring on representative new data.